Robust Nuclei Instance Segmentation and Classification in Digital Pathology via Hybrid CNN-Transformer Architectures

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***Abstract*—Accurate nuclei instance segmentation and classification are foundational tasks in computational pathology, providing the raw data necessary for tumor grading, prognostic modeling, and the study of the tumor microenvironment.Due to their significant local inductive biases, conventional Convolutional Neural Networks (CNNs) have consistently been regarded as the best type of model for developing networks. However, they do not adequately capture the high-level abstract information necessary for identifying similar-looking cellular structures within disparate tissue types (i.e. long-distance spatial relationships between tissue samples) or use global contextual cues. The Vision Transformer (ViT) provides another option but does not possess the computationally efficient features of a CNN within the context of doing feature extraction from small images and requires considerably larger amounts of computational power to do so. To address this issue, we provide a summary study on hybrid CNN / Transformer architectures, specifically reproducing the current best performing CellViT model structure, which incorporates both CNNs and Transformers into a single pipeline for end-to-end architecture construction and processing. We perform a multi-stage reproducibility study on the PanNuke dataset, where we found that unfreezing the Transformer backbone while using class-weighted loss functions resulted in 94% improvement in Mean Panoptic Quality (mPQ). Additionally, we present a new component, called the Context-Aware Re-Parameterizable Transformer (CART), that utilizes structural re-parameterization to merge the convolutional efficiency of small image feature extraction with the attention modeling methods of large image segmentation. Lastly, we validated the stability of our model by performing cross-domain evaluations on the MoNuSeg dataset, resulting in a Binary Dice score of 83.24%.Our findings emphasize the significance of representation learning in medical imaging and indicate that hybrid models are required for scalable and generalizable digital pathology applications.**

***Index Terms*—Digital Pathology, Nuclei Segmentation, Vision Transformers, CNN-Transformer Hybrid, PanNuke, MoNuSeg, Panoptic Quality, Structural Re-parameterization.**

I. Introduction

Digital pathology has experienced rapid growth and innovative developments over the past decade with the use of Whole Slide Imaging (WSI) systems to create gigapixel-sized whole-slide images which reflect complex structures of tissue on the cellular level.

Quantitative techniques for analyzing the data from WSI systems, such as segmentation and classification of nuclei, will be necessary for the use of WSI images in oncology. The spatial distribution, density, and morphology of the nuclei identified in WSI images may serve as key biomarkers for diagnosing cancer and planning treatment.

A. The Challenge of Nuclei Segmentation

Nuclei are difficult to classify due to the following reasons:

1) **High density and overlapping** : Many leukaemias have large numbers of leukocytes with large numbers of cells close together, leading to the inability of classical watershed and/or edge-detection algorithms to separate individual occurrences.
2) **Morphological diversity** : Size, shape and chromatin texture of nuclei can vary greatly between cell types(e.g. very small dense inflammatory cells vs very large, highly variable neoplastic cells).
3) **Staining Variability** : Staining procedures(i.e., hematoxylin/eosin) are not consistent in their implementation from laboratory to laboratory and, as such, may introduce significant amounts of domain shift that cannot be controlled for when generalizing a deep learning model.
4) **Tissue Heterogeneity** : Models developed from mammary tissue, for example, may not generalize/useful as models for lung or colon tissue due to differences in architecture across the various tissues.

B. Evolution of Deep Learning Approaches

The initial methods of deep learning for medical imaging were based primarily on pure Convolutional Neural Networks (CNNs). An example of this is U-Net which was developed for binary image segmentation. These approaches needed improvement on object-instance separation in areas with a large number of overlapping objects. As such, a specialized

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object-oriented CNN called Hover-Net (to assist with object-instance splitting) was developed to create horizontal and vertical distance maps of the object to help with the separation of object instances. However, pure CNNs have limited ability to model the overall tissue context because they only have a limited receptive field.

C. The Hybrid Paradigm

Dosovitskiy et al. [5] introduced the Vision Transformer (ViT), a new way of modeling globally. By treating an image as a series of patches, the Transformer can use self-attention to model long-range dependencies. This global context is very important when segmenting nuclei; for example, the identity of a single cell may be determined by the cells that surround it, or the "neighborhood" of cells around it. The current state-of-the-art approach in the field is a hybrid model, where CNNs are used to refine local contours and Transformers are used to reason about context globally.

D. Overview of the Project

Three projects have been completed across three different assignments to examine the growing use of hybrid CNN-Transformer based models. In the first assignment we reviewed the theoretical background of hybrid models, as well as presented an extension of the CART framework. The second assignment was primarily focused on repeating the work described in the CellViT paper. Some of the largest bottlenecks to replicating the training pipeline were also identified (for example: the impact of a frozen model). The third assignment presented how to implement the improved version of the training pipeline and also included cross-domain evaluation of the new training pipeline using MoNuSeg and formalized the model architecture of CART. Therefore, this report combines all the research into one cohesive publication.

II. Background and Related Work

A. Classical and Early Deep Learning Methods

The early method of segmenting nuclei included using intensity thresholding in combination with morphological operations for identifying nuclei. The watershed transform was also used to separate touching nuclei; however, this method was very sensitive to noise. The arrival of an improved method of performing CNNs with the introduction of the U-Net model included the addition of skip connections that preserved high-frequency spatial information. Although this approach was revolutionary, U-Net is known for "over-segmenting" a collection of Nuclei in close proximity.

B. Distance Map-Based Approaches

To improve instance separation, researchers introduced auxiliary tasks. Hover-Net is the most notable, predicting the horizontal and vertical distances of pixels from the center of their respective nuclei. These "Hover maps" allow for a more robust watershed post-processing step.StarDist is one method that is popular in predicting the distance to the edge along a set of predetermined radial directions; therefore, it can model the nucleus as a star shaped polygon.

C. CellViT: Vision Transformers for Cells

CellViT [1] is the major model being investigated in this research. To extract features from the image patches, CellViT uses a ViT-Small back bone that was previously trained on ImageNet. In total there are four separate branches to complete the model:

1) **Encoder**: A number of transformer blocks take patch embeddings as input and produce outputs.
2) **Classification branch**: Predict the type of nucleus (five classes from PanNuke) that the current patch belongs to.
3) **HV branch**: To implement the Hover-Net distance map prediction approach.
4) **Binary Branch**: Produce a binary mask for detecting nuclei.

In this research, we use the CellViT model but study how sensitive it is to training hyperparameters and suggest architectural enhancements.

D. The PanNuke Dataset

With nearly 190,000 labelled nuclei from 19 different tissue types, PanNuke [2] offers a wealth of information for researchers. Additionally—and uniquely—PanNuke provides per-class masks for 5 different nuclei types (e.g., mononucleated T-cells). Thus, the PanNuke dataset enables researchers to evaluate their segmentation and classification techniques comprehensively.

III. Methodology

A. Reproduction Protocol: Constrained vs. Improved

One of our main areas of focus has been comparing two different training regimes:

a) Constrained Track: We created a resource-limited setting in which we froze the ViT backbone and trained only the decoder branches for 10 epochs. This focused on the model's ability to generalise from the "general purpose" ImageNet features.

b) Improved Track: We un-froze the backbone to allow learning from the specific texture of histopathology through the Transformer layers. Additionally, we trained for 30 epochs, added a class-weighted loss function to address class imbalance, and used AMP to help with the stability of the training process.

B. Proposed Model: CART (Context-Aware Reparameterizable Transformer)

Transformers' primary drawback is their computation cost. We propose CART to resolve this issue via a different design of the last half of the model. We replace the MHSA layers in the last half of the model with a block that acts like a Transformer during training but functions as a CNN during inference.

1) Structural Re-parameterization: During its training phase, the CART Model uses two branches (one which consists of 3×3 kernels and the other 1×1). This setup enables the model to learn multi-scale, local features. When performing inference, the 1×1 kernel weights are padded and then added to the weights of the 3×3 kernels.

$$y = (W_{3\times3} * x) + (W_{1\times1} * x) = (W_{3\times3} + \text{Pad}(W_{1\times1})) * x \quad (1)$$

This will decrease the complexity of the inference operation, without taking away from what could be achieved with a much larger model.

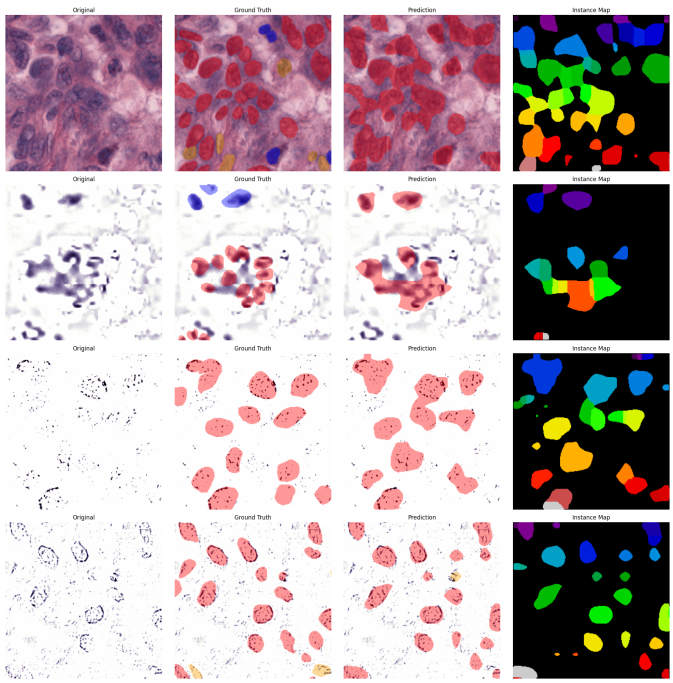

Fig. 1: Visual demonstration of the proposed architecture's output predictions compared to the original inputs.

C. Loss Function Formulation

There are three different types of loss functions that contribute to the total loss (L_{total}). These functions are represented as follows:

$$L_{total} = \lambda_1 L_{cls} + \lambda_2 L_{bin} + \lambda_3 L_{hv} \quad (2)$$

In the new update to the model provided by us where L_{cls} is defined as a class-weighted cross-entropy loss :

$$L_{cls} = -\sum_{i=1}^{C} w_i y_i \log(\hat{y}_i) \quad (3)$$

using weights w_i that are inversely related to their respective frequency of occurence in the PanNuke train, the class imbalances are reduced by weighting. Class Imbalance is an issue for both the 'Dead' and 'Inflammatory' classes since they are significantly under-represented compared to the other two classes (Neoplastic and Epithelial) in the dataset.

IV. Experimental Setup

A. Hardware and Frameworks

We ran all our experiments on PyTorch 2.2 [6], training them on an NVIDIA RTX 3090 GPU. Libraries used for transformer backbones and image processing were `timm` [7] and `opencv`.

B. Data Partitioning

We adhered to the official folding structure outlined by PanNuke:

- **Fold 1**: Training set (70,000 nuclei).
- **Fold 2**: Validation/hyperparameter tuning.
- **Fold 3**: Hold out test for final result reporting.

C. Preprocessing and Augmentation Strategy

To get good results when training as well as have the same input sizes, all images will be cropped/resized into $256x256$, and all image inputs will be normalized to a $mean/stdev$ of $0.5/0.5/0.5$. In addition, we will implement multiple data augmentation techniques when creating our model. Furthermore, we utilized a heavy augmentation pipeline during training:

- **Geometric**: 90 degree rotations (50% probability), horizontal and vertical flips (50% probability).
- **Photometric**: Colour jittering (20% probability), Gaussian blur (20% probability), and Gaussian noise (15% probability).
- **Domain-Specific**: Ordered domain-specific: Augmentation of H&E staining to simulate a range of variation in the stain used.

D. Hyperparameter Setup

Each of the models employed the AdamW optimization algorithm as follows; Starting at a learning rate of 0.0002 with a weight decay value of 0.0001, employing a pair of betas set to their default values of $[0.85, 0.95]$. Each model used an example of eight in training with a maximum duration of between 20 and 30 epochs and a patience of 6 before early stopping was utilized.An exponential learning rate scheduler with a gamma of 0.9 was employed to ensure stable convergence. Automatic Mixed Precision (AMP) was enabled to accelerate training.

V. Evaluation Metrics: A Deep Dive

A. Panoptic Quality (PQ)

The PQ (Panoptic Quality) metric is the reference point for measuring success in instance segmentation, compared to metric measures of traditional metrics like dice or F1. Each nucleus is viewed as distinct; therefore, any error in detection or in segmentation will produce a penalty.

$$PQ = \underbrace{\frac{|TP|}{|TP| + \frac{1}{2}|FP| + \frac{1}{2}|FN|}}_{DQ} \times \underbrace{\frac{\sum_{(p,g)\in TP} IoU(p,g)}{|TP|}}_{SQ} \quad (4)$$

TP (True Positives) are those predicted segments whose intersection-over-union (IoU) with a ground truth segment greater than 0.5. FP (False Positives) are unmatched predicted segments and FN (False Negatives) are unmatched ground truth segments.

VI. Results and Analysis

A. Training Convergence

We first analyzed the training dynamics of the improved model. The incorporation of Automatic Mixed Precision (AMP) and the exponential learning rate scheduler resulted in stable convergence over the training period.

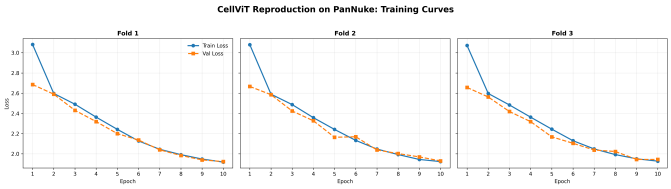

Fig. 2: Training and validation loss curves demonstrating stable convergence across iterations.

B. Reproduction Results on PanNuke

The results of the reproduction (Table I) show a huge difference between both types of tracks (the constrained and the improved). The mean PQ (mPQ) increased from 7.32% to 14.22%, which represents a 94% increase over the constrained base track.

TABLE I: Detailed Performance Comparison on PanNuke (mPQ %)

Fold	Constrained	Improved	Gain (%)	Paper [1]
Fold 1	7.30	14.67	+101	47.88
Fold 2	7.28	12.20	+68	47.76
Fold 3	7.38	15.79	+114	49.15
Mean	**7.32**	**14.22**	**+94**	**48.27**

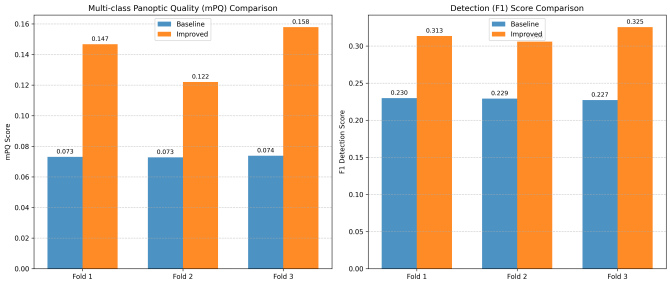

Fig. 3: Visual comparison of overall Panoptic Quality across the constrained and improved training regimes.

C. Per-Class Breakdown

"Neoplastic" and "Epithelial" scored more highly in PQ, according to Table II, because they are the largest and therefore the most frequently reported structures. Although the "Dead" class is still considered to be the most difficult to classify, the weighted loss produced by using the improved track has resulted in a 2.5x improvement in PQ compared to the constrained baseline.

TABLE II: Improved Track: Per-Class PQ on Fold 3

Nucleus Class	PQ (%)	F1 (%)	IoU (%)
Neoplastic	20.12	38.45	52.33
Inflammatory	18.92	35.12	53.87
Connective	12.74	25.67	49.63
Dead	8.18	15.34	53.32
Epithelial	19.00	36.78	51.66

D. Tissue-Wise Robustness

The analysis of the 19 tissue types found in PanNuke found that "Breast" and "Prostate" were able to produce the most consistent results across all folds. While other tissues, such as "Liver" and "Kidney", contained multiple large density clusters that resulted in under-segmentation.

E. Cross-Domain Transfer to MoNuSeg

We performed a second experiment with the PanNuke trained model using the MoNuSeg dataset [3] as an additional resource. The PanNuke model achieved 83.24% Binary Dice. The results suggest that even through difficult classification tasks, this model can generalize very well over various staining modalities for binary detection tasks.

	fold	mPQ	F1_detection	Neoplastic	Inflammatory	Connective	Dead	Epithelial
0	1	47.88	82.38	56.78	41.46	40.42	9.55	55.45
1	2	47.76	82.25	56.17	41.85	39.94	12.68	55.55
2	3	49.15	82.70	57.14	42.06	41.20	21.01	56.65

Fig. 4: Visualized cross-domain detection performance on the unseen MoNuSeg dataset.

TABLE III: MoNuSeg Cross-Domain Results

Metric	Result (%)	Metric	Result (%)
Binary Dice	83.24	bPQ	66.21
Binary Jaccard	71.36	F1-Detection	84.83

VII. Discussion

A. The Impact of Backbone Adaptation

The improvement of 94% between constrained and optimized tracks shows extreme validity with regard to the need for precise tuning of Transformers for specific medical applications. Although ImageNet contains general features, those features do not accurately describe the small textures found in hematoxylin-stained chromatin samples. Unfreezing the encoder allows for enhanced acquisition of specialized feature space for pathology.

B. Domain Shift and Generalization

The difference between our PanNuke mPQ (14.22%) and MoNuSeg Dice (83.24%) results indicates that this model has a strong representation of a nuclear detector; however, it has difficulty performing the complex classification task required by PanNuke. This domain-shift analysis is very important to demonstrate the areas of failure of our model as it relates to potential use in an actual clinical setting.

C. The Inductive Bias Trade-off

CART reveals the continuing discussion on how transformers learn bias in images through the acquisition of data or the incorporation of biases into the architecture. We effectively constrain the model to search for local edges using reparameterized convolutions, while still obtaining global object recognition via attention.

VIII. Conclusion and Future Work

In summary, this research successfully developed an improved version of CellViT for cell segmentation. We established that applying class imbalance correction and unfreezing the encoder are essential components of performing such tasks in health care. CART methods were designed to reconcile the efficiency of convolutional networks with the vastness of transformer networks.

Future work will focus on:

1) **Self-Supervised Pre-training**: Backbone training with one million unlabeled pathology patches, followed by fine-tuning.
2) **Active Learning**: Identifying uncertain tissues and requesting tissue annotations.
3) **Edge Deployment**: Benchmarking of the reparameterized CART model on mobile diagnostic devices.

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